

Propositional Inference: Logical Equivalence, Truth Tables, and Model Checking

Lecture 14

Last Time: What Does Entailment Mean?

A knowledge base constrains the set of worlds the agent considers possible.

$$KB \models \alpha$$

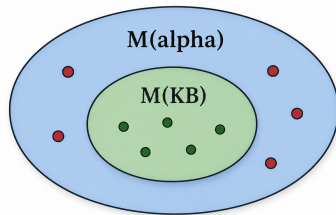
means that **every model satisfying the knowledge base also satisfies α** .

The missing piece

The definition tells us what entailment *means*, but not yet how an agent can decide it computationally.

Key idea

Today we turn the semantic definition of entailment into an inference procedure.



$M(KB) \text{ subset } M(\alpha)$

The Logical Agent Has a Query

Suppose a delivery robot knows

$$B \rightarrow C$$

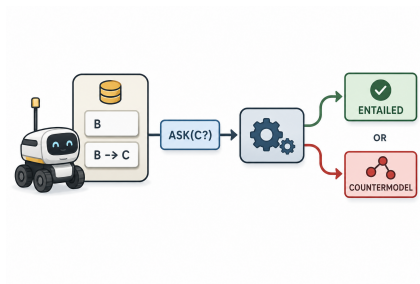
$$B$$

where B means “battery is low” and C means “robot should charge.”

The agent asks:

$$KB \models C?$$

We need a procedure that either confirms the query or finds a counterexample world.



Entailment Is a Universal Claim

- ▶ $KB \models \alpha$ says something about **every** model satisfying KB .
- ▶ To prove entailment semantically, all KB -models must make α true.
- ▶ To disprove entailment, one countermodel is enough.

Key idea

Entailment is “for all surviving worlds”; non-entailment needs only one surviving world where the query is false.

Countermodels: The Fastest Way to Break a Claim

Consider the claim

$$P \vee Q \models P.$$

Try the model

$$P = \text{False}, \quad Q = \text{True}.$$

Then $P \vee Q$ is true while P is false.

Conclusion

That single model disproves the entailment.

$P = \text{F}$ $Q = \text{T}$
$P \text{ OR } Q = \text{T}$ ✓
Query $P = \text{F}$ ✗

COUNTERMODEL

Three Useful Semantic Categories

Tautology: a sentence true in every model.

Contradiction: a sentence false in every model.

Satisfiable: a sentence true in at least one model.

Important distinction

A sentence can be satisfiable without being a tautology.

Key idea

These are properties of a sentence across its possible models.

Truth Tables Enumerate Possible Worlds

P	Q
T	T
T	F
F	T
F	F

With two proposition symbols there are

$$2^2 = 4$$

complete models.

Truth Tables Give the Meaning of Connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

Key idea

A truth table is a complete semantic specification of a propositional connective.

Implication Revisited

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

$P \rightarrow Q$ is false only when P is true and Q is false.

Key idea

Implication is a constraint: the forbidden world is $P = T, Q = F$.

Biconditional Means “Same Truth Value”

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

$P \leftrightarrow Q$ is true exactly when P and Q agree.

Logical Equivalence

Definition

Two sentences are **logically equivalent** when they have the same truth value in every model.

We write

$$\alpha \equiv \beta.$$

Equivalence allows us to replace one expression with another without changing its meaning.

Key idea

Logical equivalence is semantic sameness, not merely visual similarity.

Example: Eliminate an Implication

P	Q	$P \rightarrow Q$	$\neg P \vee Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

The final two columns match in every row, so

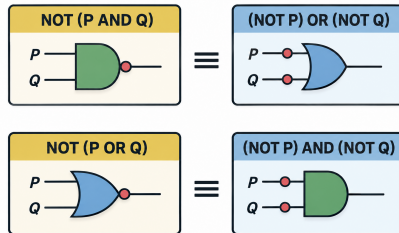
$$P \rightarrow Q \equiv \neg P \vee Q.$$

De Morgan's Laws

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Negating a conjunction changes AND to OR; negating a disjunction changes OR to AND.



Other Equivalences Worth Knowing

- ▶ Double negation: $\neg\neg P \equiv P$.
- ▶ Commutativity: $P \wedge Q \equiv Q \wedge P$ and $P \vee Q \equiv Q \vee P$.
- ▶ Associativity lets repeated ANDs or ORs regroup.
- ▶ Distributivity lets AND and OR move across one another.
- ▶ Biconditional:

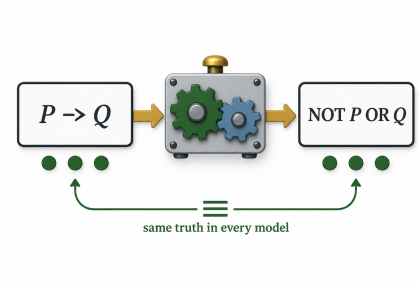
$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P).$$

Key idea

These identities become tools for automated rewriting next lecture.

Why Equivalence Matters to an AI System

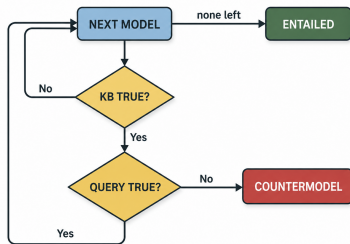
- ▶ Different-looking sentences can encode exactly the same information.
- ▶ A reasoning system can rewrite expressions into forms that are easier to process.
- ▶ A valid rewrite must preserve the set of satisfying models.
- ▶ Next lecture, we exploit this idea with normal forms.



Model Checking: Use the Definition Directly

To test $KB \models \alpha$:

1. Enumerate all possible models.
2. Ignore models where KB is false.
3. Inspect α in every model where KB is true.
4. If α is true in all of them, entailment holds.



Worked Example: Does the KB Entail Q ?

$$KB = (P \rightarrow Q) \wedge P$$

Query: Q

We will evaluate the knowledge base and query across all four models for P and Q .

Step 1: Enumerate the Models

P	Q	$P \rightarrow Q$	$KB = (P \rightarrow Q) \wedge P$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

Only rows where the KB column is true matter.

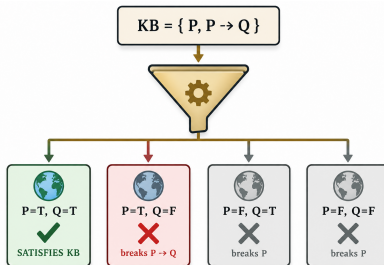
Step 2: Keep Only *KB*-Models

The knowledge base is true only in the model

$$P = T, \quad Q = T.$$

Every other model violates at least one sentence in the knowledge base.

The KB has narrowed four possible worlds down to one.



Step 3: Inspect the Query in Every KB -Model

In the only surviving model, Q is true.

There is no model where the knowledge base is true and Q is false.

Therefore

$$KB \models Q.$$

Key idea

The conclusion follows because the knowledge base leaves no possible world in which the query is false.

A Non-Entailment Example

Let

$$KB = P \vee Q$$

with query P .

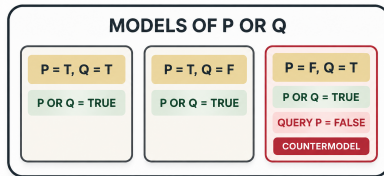
The model

$$P = F, \quad Q = T$$

satisfies the knowledge base but falsifies the query.

Therefore

$$P \vee Q \not\models P.$$



Truth-Table Entailment as an Algorithm

TT-ENTAILS?(KB, α)

```
for every complete truth assignment  $m$ :  
  if  $m$  satisfies  $KB$ :  
    if  $m$  does not satisfy  $\alpha$ : return false  
return true
```

Key idea

This is exhaustive model checking: search all possible worlds for a countermodel.

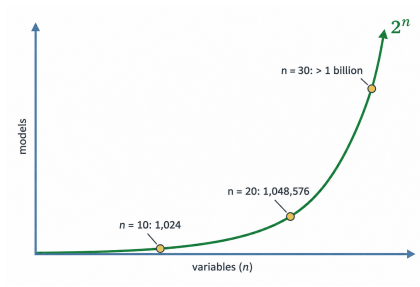
The Cost: 2^n Models

With n proposition symbols, there are

$$2^n$$

complete truth assignments.

- ▶ 10 symbols $\rightarrow$ 1,024 models.
- ▶ 20 symbols $\rightarrow$ 1,048,576 models.
- ▶ 30 symbols $\rightarrow$ more than one billion models.

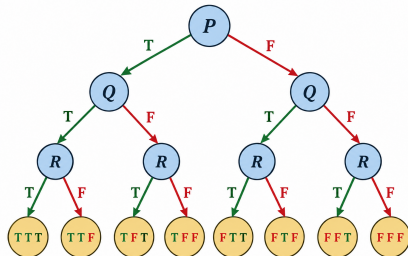


Key idea

The semantics are simple; exhaustive inference scales badly.

This Is Another Search Problem

- ▶ **State:** a partial or complete truth assignment.
- ▶ **Branching choice:** assign the next proposition true or false.
- ▶ **Target counterexample:** a model satisfying KB and falsifying the query.
- ▶ Truth-table checking explores the complete binary assignment tree.



Key idea

A recurring AI theme: reasoning can often be understood as search over a structured space.

Entailment vs. Inference

Semantic relation

$$KB \models \alpha$$

α is true in every model of KB .

Syntactic derivation

$$KB \vdash_i \alpha$$

Inference procedure i derives α from KB .

Key idea

Do not confuse “it follows” with “this particular procedure found it.”

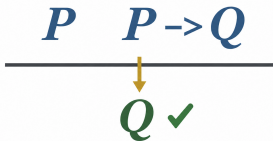
Inference Rules: Reason Without Enumerating Every World

An inference rule licenses a new sentence from existing sentences.

Example: Modus Ponens

$$\frac{P \quad P \rightarrow Q}{Q}$$

This is a syntactic operation, but we want it to preserve semantic truth.



A diagram illustrating the Modus Ponens inference rule. It shows the premises P and $P \rightarrow Q$ above a horizontal line. A yellow arrow points from the line down to the conclusion Q , which is followed by a green checkmark, indicating that the rule preserves semantic truth.

Modus Ponens Is Sound

$$\begin{array}{c} P \\ P \rightarrow Q \\ \hline \therefore Q \end{array}$$

If both premises are true, there is no model in which the conclusion is false.

Key idea

Modus Ponens never manufactures a false conclusion from true premises.

A Tempting but Invalid Rule

Suppose

$$P \rightarrow Q$$

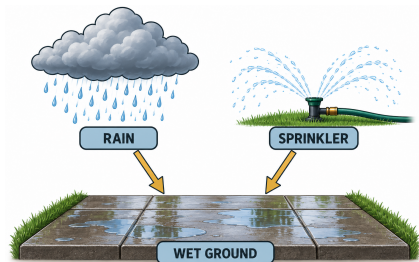
and

$$Q.$$

Can we conclude P ? **No.**

Wet ground does not imply rain; a sprinkler may also explain it.

This is the fallacy of **affirming the consequent**.



Soundness

Definition

An inference procedure is **sound** if every sentence it derives is actually entailed.

$$KB \vdash_i \alpha \Rightarrow KB \models \alpha$$

A sound procedure may fail to find some truths, but it will not derive unsupported conclusions.

Key idea

Soundness protects correctness.

Completeness

Definition

An inference procedure is **complete** if it can derive every entailed sentence.

$$KB \models \alpha \Rightarrow KB \vdash_i \alpha$$

A complete procedure cannot permanently miss a conclusion that really follows.

Key idea

Completeness protects coverage.

Truth-Table Model Checking Is Sound and Complete

- ▶ It checks the semantic definition directly.
- ▶ If it says “entailed,” there is no countermodel.
- ▶ If entailment holds, exhaustive enumeration will confirm it.
- ▶ The problem is not correctness—it is cost.

Key idea

We now have a correct baseline algorithm. The next problem is doing less work.

Satisfiability Gives Us Another View of Entailment

$KB \models \alpha$ exactly when there is no model satisfying KB while α is false.

That means

$$KB \wedge \neg \alpha$$

must be unsatisfiable.

Therefore

$$KB \models \alpha \iff KB \wedge \neg \alpha \text{ is unsatisfiable.}$$

Key idea

To prove α , assume $\neg \alpha$ and show that no consistent world remains.

Example: Turn Entailment into a Contradiction Test

Want to prove

$$\{P, P \rightarrow Q\} \models Q.$$

Add the negation of the query:

$$\neg Q.$$

Now ask whether

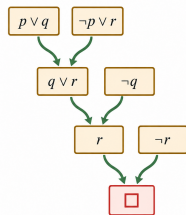
$$P \wedge (P \rightarrow Q) \wedge \neg Q$$

can be satisfied.

It cannot: P and $P \rightarrow Q$ force Q , while $\neg Q$ forces Q to be false.

Why This Reformulation Matters

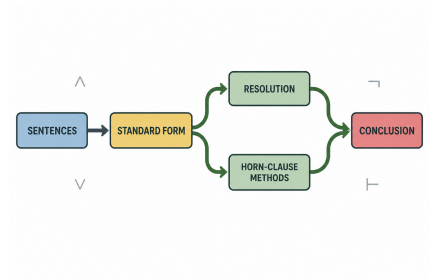
- ▶ Truth tables prove entailment by checking all possible worlds.
- ▶ Unsatisfiability reframes the same problem as deriving an inconsistency.
- ▶ Perhaps we can manipulate sentences directly instead of explicitly enumerating every model.



Next: Automated Logical Inference

Next lecture we will

- ▶ convert sentences into a standardized form;
- ▶ use resolution to derive contradictions mechanically;
- ▶ exploit Horn-clause structure;
- ▶ compare forward and backward chaining.



Takeaways

- ▶ Truth tables enumerate models and can decide propositional entailment.
- ▶ Logical equivalence means identical truth values in every model.
- ▶ A single countermodel disproves entailment.
- ▶ Soundness protects correctness; completeness protects coverage.
- ▶ $KB \models \alpha$ iff $KB \wedge \neg\alpha$ is unsatisfiable.
- ▶ Next lecture: reason by manipulating sentences rather than enumerating worlds.

Key idea

Semantic inference gives us a correct baseline; automated proof procedures aim to keep the guarantees while doing less work.